Tribhuvan University

Faculty of Science and Technology

OFFICE OF THE DEAN

2079

Bachelor of Science in Computer Science and Information Technology

Course Title: **Advanced Database**

Code No: **CSC461** Semester: **8th Semester**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Section A

1. Design an EER model for library management system having generalization and specialization hierarchies. The EER model should have disjoint and overlapping constraints with at least one of the entities having total participation. Use your own assumption for another concepts. Now convert so designed EER diagram into its equivalent relational model.

2. What are different concepts and features of object-oriented database? What is object relational model?

3. Define distributed database. What are the benefits of using distributed database over centralized database? Explain availability, reliability and scalability features of distributed databases.

Section B

Attempt any Eight questions[8×5=40]

4. Why important to store data in database? What is multilevel index?

5. What is aggregation? Explain with example.

6. Define ODMG object model. What is object Query language (OQL)?

7. What is query processing? Differentiate query processing with query optimization.

8. Why query optimization is necessary? Illustrate on the choice of query execution plans.

9. Why fragmentation is carried out? Differentiate between horizontal and vertical fragmentation.

10. Why NOSQL system is essential? What are different characteristics of this system?

11. What is spatial database? Explain the concept of trigger with an example.

12. Write short notes ona. CAP Theoremb. Deductive database